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# SPECIALIZATION IN PYRAMIDAL CELL STRUCTURE IN THE SENSORY-MOTOR CORTEX OF THE VERVET MONKEY (*CERCOPETHICUS PYGERYTHRUS*)

**G. N. ELSTON,[a]* R. BENAVIDES-PICCIONE,[b] A. ELSTON,[a] J. DEFELIPE,[b] AND P. R. MANGER[c]**

[a]*Vision, Touch and Hearing Research Centre, Department of Physiology and Pharmacology, School of Biomedical Sciences and Queensland Brain Institute, the University of Queensland, Queensland, 4072, Australia*

[b]*Instituto Cajal (CSIC), Avda Dr Arce, 37, 28002, Madrid, Spain*

[c]*School of Anatomical Sciences, Faculty of Health Sciences, University of the Witwatersrand, Johannesburg, Parktown, 2193, South Africa*

**Abstract—Recent studies have revealed systematic differences in the pyramidal cell structure between functionally related cortical areas of primates. Trends for a parallel in pyramidal cell structure and functional complexity have been reported in visual, somatosensory, motor, cingulate and prefrontal cortex in the macaque monkey cortex. These specializations in structure have been interpreted as being fundamental in determining cellular and systems function, endowing circuits in these different cortical areas with different computational power. In the present study we extend our initial finding of systematic specialization of pyramidal cell structure in sensory-motor cortex in the macaque monkey [Cereb Cortex 12 (2002) 1071] to the vervet monkey. More specifically, we investigated pyramidal cell structure in somatosensory and motor areas 1/2, 5, 7, 4 and 6. Neurones in fixed, flat-mounted, cortical slices were injected intracellularly with Lucifer Yellow and processed for a light-stable 3,3′-diaminobenzidine reaction product. The size of, number of branches in, and spine density of the basal dendritic arbors varied systematically such that there was a trend for increasing complexity in arbor structure with progression through 1/2, 5 and 7. In addition, cells in area 6 were larger, more branched, and more spinous than those in area 4.** **© 2005 IBRO. Published by Elsevier Ltd. All rights reserved.**

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Pyramidal cell structure has been shown to differ between cortical areas involved in somatosensory and motor processing (Cajal, 1899; Conel, 1941, 1947, 1955, 1959, 1963, 1967; Hendry and Jones, 1983; Lund et al., 1993; Elston and Rockland, 2002). Moreover, the regional specialization in pyramidal cell structure is systematic: pyramidal cells in the primary somatosensory area are smaller, less branched and less spinous than those in Brodmann's area 5, which are smaller, less branched and less spinous than those in Brodmann's area 7 (Elston and Rockland, 2002). Those in primary motor cortex (Brodmann's area 4) are smaller, less branched and less spinous than those in premotor cortex (Brodmann's area 6). It has been suggested that as pyramidal cells are the most common neuronal type in cerebral cortex, specialization in cell structure reflects specialization in cortical circuitry. More specifically, circuitry in cortical areas composed of neurones that integrate more inputs is likely to be more complex than that composed of neurones that integrate relatively few inputs (see Elston, 2002, 2003a; Jacobs and Scheibel, 2002 for reviews). There have been relatively few studies of pyramidal cell structure in sensory-motor cortex and the published data on pyramidal cell morphology in the tangential plane comes only from the macaque monkey (Elston and Rockland, 2002). Consequently, it remains unknown as to how circuitry in sensory-motor cortex may differ between species, how expansion of the parietal lobe (Brodmann, 1909, 1913; see Elston and Garey, 2004; Garey, 1994 for translations) may have influenced the evolution of circuitry in sensory-motor cortex and how that may parallel sensory-motor function in different primates (e.g. knuckle walking vs. upright gait). In the present investigation we extend the basis for comparison by studying pyramidal cell structure in the sensory-motor cortex of the vervet monkey, a cercopithecid monkey also known as the savannah guenon that diverged from the present day macaque monkeys approximately 10 million years ago (Dawkins, 2004).

## EXPERIMENTAL PROCEDURES

### Tissue preparation

Two adult male vervet monkeys (VM1 and VM2), also know as the savannah guenon (*Cercopithecus pygerythrus*), of similar weight (6.1 and 5.6 kg, respectively) were overdosed with sodium pentobarbitol, perfused intracardially with phosphate-buffered saline, then 4% paraformaldehyde in 0.1 mol/l phosphate buffer (PB) (pH 7.2) and their brains removed. Tissue was taken from the precentral gyrus medial to the superior precentral dimple (area 6 of Brodmann, 1909), the exposed portion of the precentral gyrus immediately anterior to the central sulcus (area 4 of Brodmann, 1909), the postcentral gyrus immediately posterior to the central sulcus (areas 1/2 of Vogt and Vogt, 1919; Powell and Mountcastle, 1959a,b; Sur et al., 1980; Pons et al., 1985), the exposed bank of the posterior postcentral gyrus (Brodmann's area 5) and the intraparietal gyrus (area 7b of Vogt and Vogt, 1919) (Fig. 1). As in a previous study (Elston and Rockland, 2002) we sampled tissue along a rostrocaudal transect through all areas (Fig. 2) in a bid to increase the likelihood that we studied cells from similar body part representations. Based on what is known of the topography in other primates (Merzenich et al., 1978, 1987; Matsumura

*Corresponding author. Tel: +61-7-3365-4484; fax: +61-7-3365-4522.
E-mail address: G.Elston@vthrc.uq.edu.au (G. Elston).
*Abbreviations:* ANOVA, analysis of variance; DAB, 3,3′-diaminobenzidine; DAPI, 4,6 diamidino-2-phenylindole; PB, phosphate buffer.

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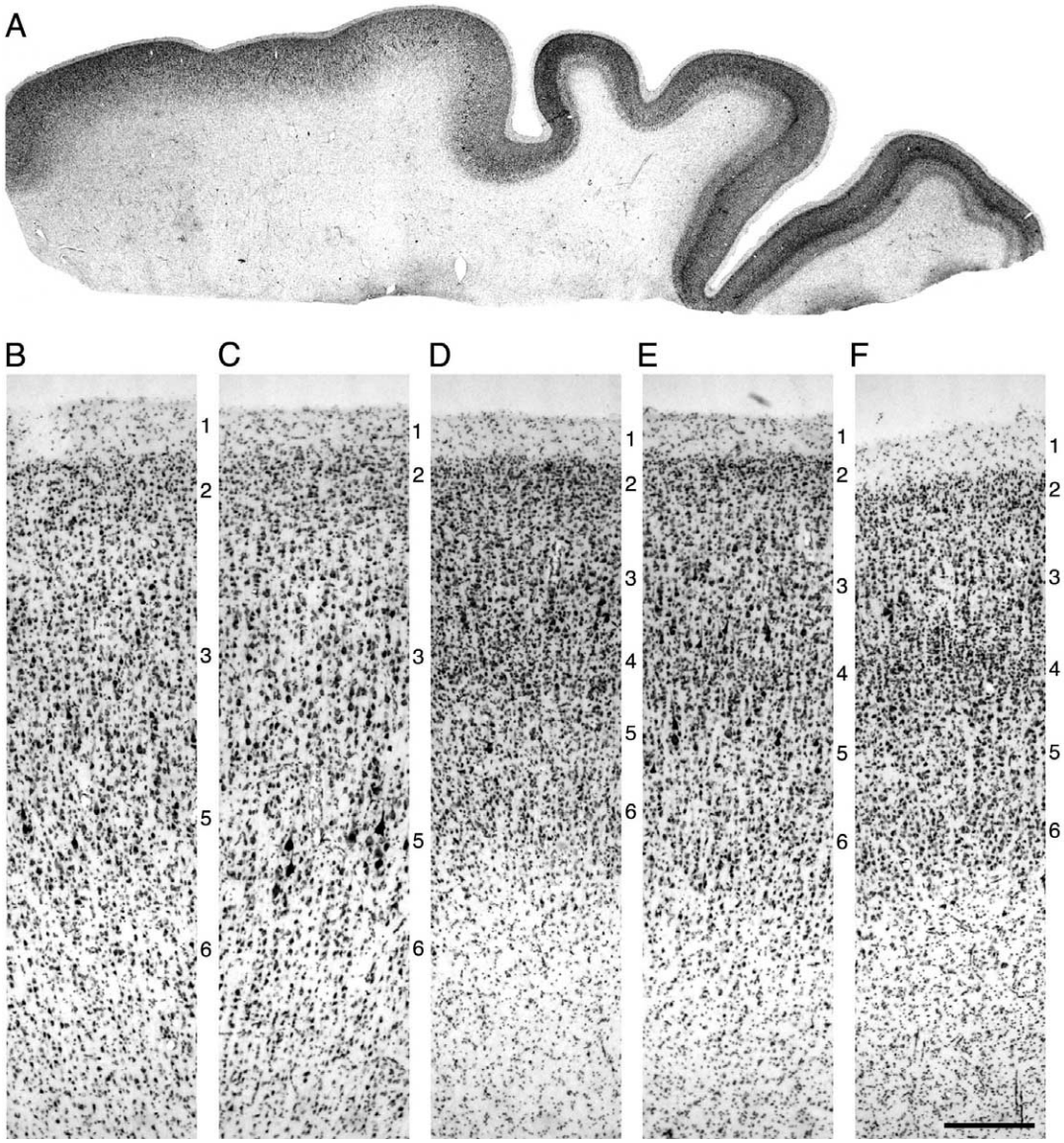



**Fig. 1.** Photomicrographs of (A) of a single transverse section illustrating cortical areas from which neurones were sampled. The section transverses from the arcuate sulcus to the parieto-occipital sulcus, parallel to the midline. Based on differences in cytoarchitecture it is possible to distinguish with certainty the border between the granular primary somatosensory area 3b and the non-granular primary motor area (area 4) and between area 4 and premotor cortex (area 6). In addition, it is possible to distinguish the border between area 3b and area 1. (B–F) Higher power photomicrographs of (B) area 6, (C) area 4, (D) area 1/2, (E) area 5 and (F) area 7 illustrating differences in lamination patterns. Scale bar=250 μm in B–F.

and Kubota, 1979; Pons and Kaas, 1986; Manger et al., 1996; Wu and Kaas, 2003), it is likely that cells in the present study were sampled in the hindlimb representation. Tissue was then postfixed overnight in 4% paraformaldehyde in 0.1 mol/l PB as flat-mounts. The next day, individual sections (250 μm) were cut tangential to the cortical surface with the aid of a vibratome. All tissue was sampled from the left hemisphere.

## Intracellular injection

Sections were prelabeled with 4,6 diamidino-2-phenylindole (DAPI; Sigma, IL, USA, D9542), which fluoresces under UV excitation (see Fig. 3 of Elston and Rosa, 1997) enabling injection of individually labeled cell bodies under visual control (see Buhl and Schlote, 1987; Einstein, 1988; Elston et al., 1997; Elston and Rosa, 1997 for details). Cytoarchitecture was revealed by both the fluorescent DAPI label in which differences in neuronal packing density and type are easily detectable (see Fig. 3 of Elston and Rosa, 1997) and from preparations stained for Nissl-substance (Fig. 1). In somatosensory cortex, layer III was identified as that immediately superior to the cell dense granular layer IV. In motor cortex, the location of the base of layer III was estimated from the transition in neuronal density and soma size in the DAPI- and Nissl-labeled preparations (see Fig. 1). Pyramidal neurones were injected by continuous current (up to 100 nA) until the individual dendrites of each cell could be traced to abrupt distal tips and the dendritic spines were easily visible. The sections were then processed with 1:400 000 anti-Lucifer Yellow in 2% bovine serum albumin (Sigma A3425), 1% Triton X-100 (BDH 30632) and 5% sucrose in 0.1 mol/l PB for 5 days at room temperature, washed three times in PB, then processed with 1:200 Amersham biotinylated anti-rabbit (Amersham RPN 1004) in 2% bovine serum albumin, 1%

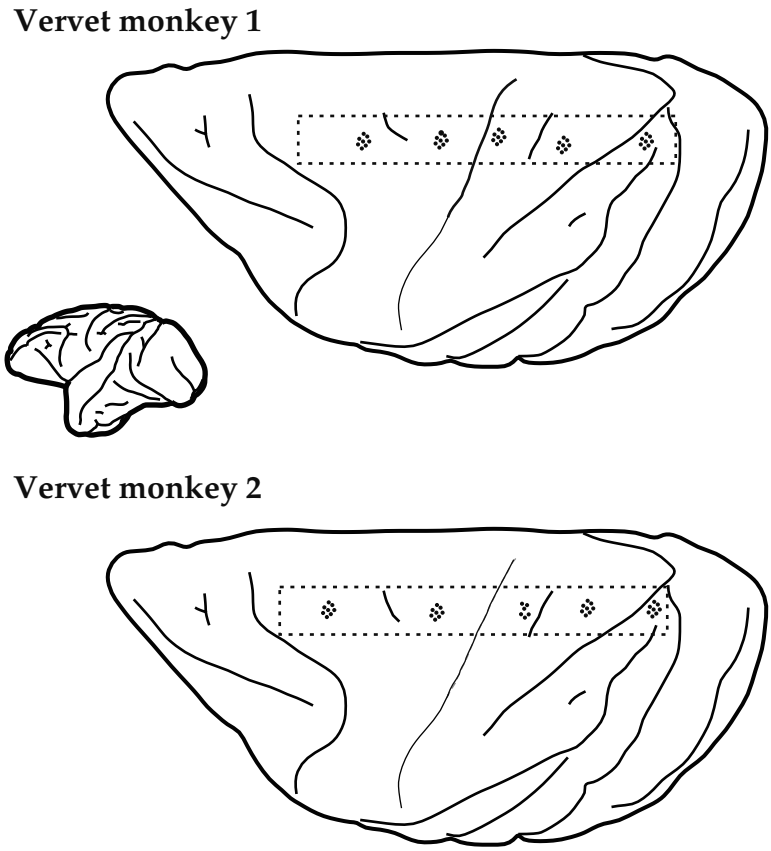



**Fig. 2.** Schematic of the dorsal aspect of the vervet monkey cerebral cortex showing the regions of cortex in which neurones were injected (solid dots). The central sulcus has been "opened" to illustrate both the anterior and posterior banks (light gray shade), which contained the primary motor area (Brodmann's area 4) and the primary somatosensory area (Vogt and Vogt's area 1/2), respectively. Inset is the lateral view of the same brain.

Triton X-100 and 5% sucrose in 0.1 mol/l in PB for 2 h, washed another three times before being processed for a further 2 h in 1:200 streptivadin biotin horseradish peroxidase (Amersham RPN1051) in 0.1 mol/l PB. DAB (3,3′-diaminobenzidine; Sigma D 8001) was used as the chromogen to reveal a light-stable robust reaction product (Fig. 3).

## Morphological analyses

Two-dimensional analysis of cell structure was performed on camera lucida reconstructions of individual neurones. Only spiny cells that had an unambiguous apical dendrite were included for analyses. Only dendrites that issued from the cell body were reconstructed, apical collaterals were excluded. The size of the basal dendritic arbor was determined for each cell by scanning the drawings and calculating the areas contained within a polygon that joined the outermost distal tips of the dendrites using NIH image software (NIH Research Services, Bethesda, MD, USA). Sholl analyses (Sholl, 1953) were performed to quantify the branching structure of the basal dendrites. Spines were drawn along the entire length of 20 horizontally-projecting dendrites of different

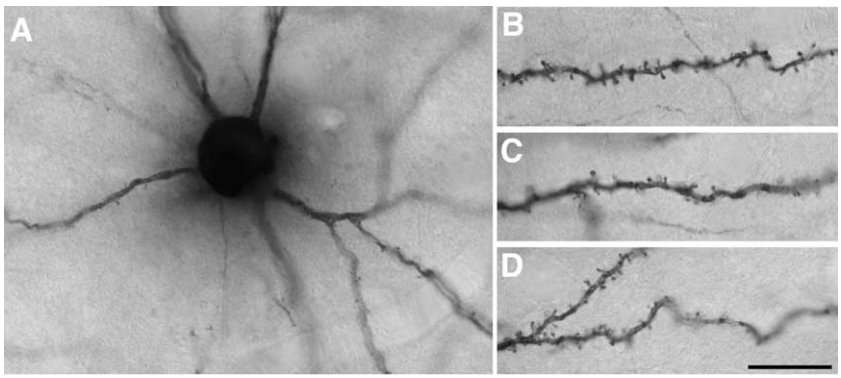



**Fig. 3.** Photomicrographs of the (A) cell body and proximal basal dendrites, and (B–D) middle region of selected basal dendrites, of layer III pyramidal cells that were injected with Lucifer Yellow and processed for a DAB reaction product. Cells were sampled from area 6, premotor cortex. Scale bar=20 μm.

**Table 1.** Basal dendritic field area of layer III pyramidal cells

| Cortical area | *n* | Mean $\mu m^2 \times 10^4$ | S.D.$\times 10^4$ | S.E.M.$\times 10^4$ | Minimum $\mu m^2 \times 10^4$ | Maximum $\mu m^2 \times 10^4$ |
|---|---|---|---|---|---|---|
| VM1 | | | | | | |
| 7 | 26 | 11.28 | 1.81 | 0.36 | 7.78 | 14.81 |
| 5 | 31 | 11.19 | 2.41 | 0.43 | 6.41 | 15.84 |
| 1/2 | 9 | 6.82 | 2.73 | 0.91 | 3.08 | 12.59 |
| 4 | 28 | 12.50 | 3.55 | 0.67 | 6.66 | 20.74 |
| 6 | 33 | 13.31 | 2.76 | 0.48 | 9.08 | 21.71 |
| VM2 | | | | | | |
| 7 | 25 | 14.32 | 2.27 | 0.45 | 9.18 | 20.14 |
| 5 | 24 | 11.40 | 1.79 | 0.37 | 8.18 | 14.81 |
| 1/2 | 38 | 10.69 | 1.98 | 0.32 | 7.40 | 16.87 |
| 4 | 19 | 12.48 | 3.30 | 0.76 | 7.14 | 20.83 |
| 6 | 41 | 13.39 | 3.16 | 0.49 | 7.58 | 19.69 |

neurones in each cortical area with the aid of a ×100 Zeiss oil immersion objective (numerical aperture=1.25). All statistical analysis was performed with SPSS (SPSS Inc., IL, USA). All experiments were performed in accordance with guidelines endorsed by the NIH (publication No. 86–23, revised 1985) and were approved by the University of the Witwatersrand Animal Ethics Committee. Every effort was made to minimize suffering and the number of animals used.

## RESULTS

Four hundred and eighty-nine neurons were injected in layer III of cortical areas 1/2, 4, 5, 6, and 7. Two hundred and seventy-three of these were included for analysis as they had an unambiguous apical dendrite characteristic of pyramidal neurones (see Feldman, 1984; DeFelipe and Fariñas, 1992 for reviews), had their complete basal dendritic arbors included within the section, and were well filled (127 in VM1 and 146 in VM2; see Table 1). The data are reported according to the animal from which it was sampled.

### Basal dendritic field areas

In both cases we observed qualitative differences in the size of the basal dendritic trees of layer III pyramidal cells in the different cortical areas. Quantification of the dendritic trees revealed a consistent trend in both animals: cells in areas 4 and 6 were of similar size, cells became progressively larger through somatosensory areas 1/2, 5 and 7, and cells in area 4 were larger than those in area 1/2 (Figs. 4A and 5A, Table 1). Analyses of variance (ANOVAs) revealed these differences to be statistically significant ($P<0.001$) in both VM1 ($F_{(4)}=11.19$) and VM2 ($F_{(4)}=10.18$). Post hoc one way ANOVAs revealed no significant difference in the size of the trees between cells in areas 4 and 6 (VM1: $F_{(1)}=0.99$; VM2: $F_{(1)}=1.13$). In VM1 there was a significant difference ($P<0.05$) in size of the dendritic trees of cells between areas 1/2 and 5 ($F_{(1)}=21.7$), but not in VM2 ($F_{(1)}=2.06$). In VM2 there was a significant difference ($P<0.05$) in size of the dendritic trees of cells between areas 5 and 7 ($F_{(1)}=24.79$), but not in VM1 ($F_{(1)}=0.026$). In both VM1 and VM2 there was a significant difference in the size of the dendritic trees of pyramidal cells between areas 3 and 4 ($F_{(1)}=19.27$ and $F_{(1)}=6.55$, respectively).

### Branching patterns of the basal dendritic arbors

Branching patterns of the basal dendritic arbors of pyramidal neurones are shown in Figs. 4B and 5B, where the results of Sholl analysis (based on concentric circles of increasing radii: 25 μm intervals) were plotted. As can be seen from the figures, the branching pattern of the basal dendritic trees of pyramidal neurones differed between all cortical areas. In both VM1 and VM2, cells in area 6 were more branched than those in area 4, cells became progressively more branched through somatosensory areas 1/2, 5 and 7 and cells in area 4 were more branched than those in area 1/2 (Figs. 4B and 5B, Table 2). In addition, the peak branching complexity in the basal dendritic trees of all pyramidal cells sampled was located approximately one-third the distance from the cell body to the distal tips of the dendrites. In order to determine whether these differences in branching patterns were statistically significant, we performed repeated measures ANOVAs (cortical area by distance from the soma, 5×11 design). These analyses revealed a significant difference ($P<0.001$) in the branching patterns of neurones in both VM1 ($F_{(1,\ 4)}=17.95$) and VM2 ($F_{(1,4)}=20.24$). Post hoc repeated measures ANOVAs revealed that cells in area 6 were significantly more branched than those in area 4 (VM1: $F_{(1,\ 1)}=12.83$; VM2: $F_{(1,\ 1)}=5.48$), and that cells in area 1/2 were significantly less branched ($P<0.05$) than those in area 5 (VM1: $F_{(1,\ 1)}=28.9$; VM2: $F_{(1,\ 1)}=3.99$), which, in turn, were significantly less branched than those in area 7 (VM1: $F_{(1,\ 1)}=11.37$; VM2: $F_{(1,\ 1)}=44.59$). Cells in area 4 were significantly more branched than those in area 1/2 (VM1: $F_{(1,\ 1)}=19.13$; VM2: $F_{(1,\ 1)}=8.63$).

### Spine densities of the basal dendrites

Over 12,000 individual spines were drawn and tallied as a function of distance along the basal dendrites of layer III pyramidal cells (5670 in VM1 and 6584 in VM2). As reported elsewhere (Elston and DeFelipe, 2002) the density of spines was low in the proximal basal dendrites, rose to a peak approximately one-third the distance from the cell

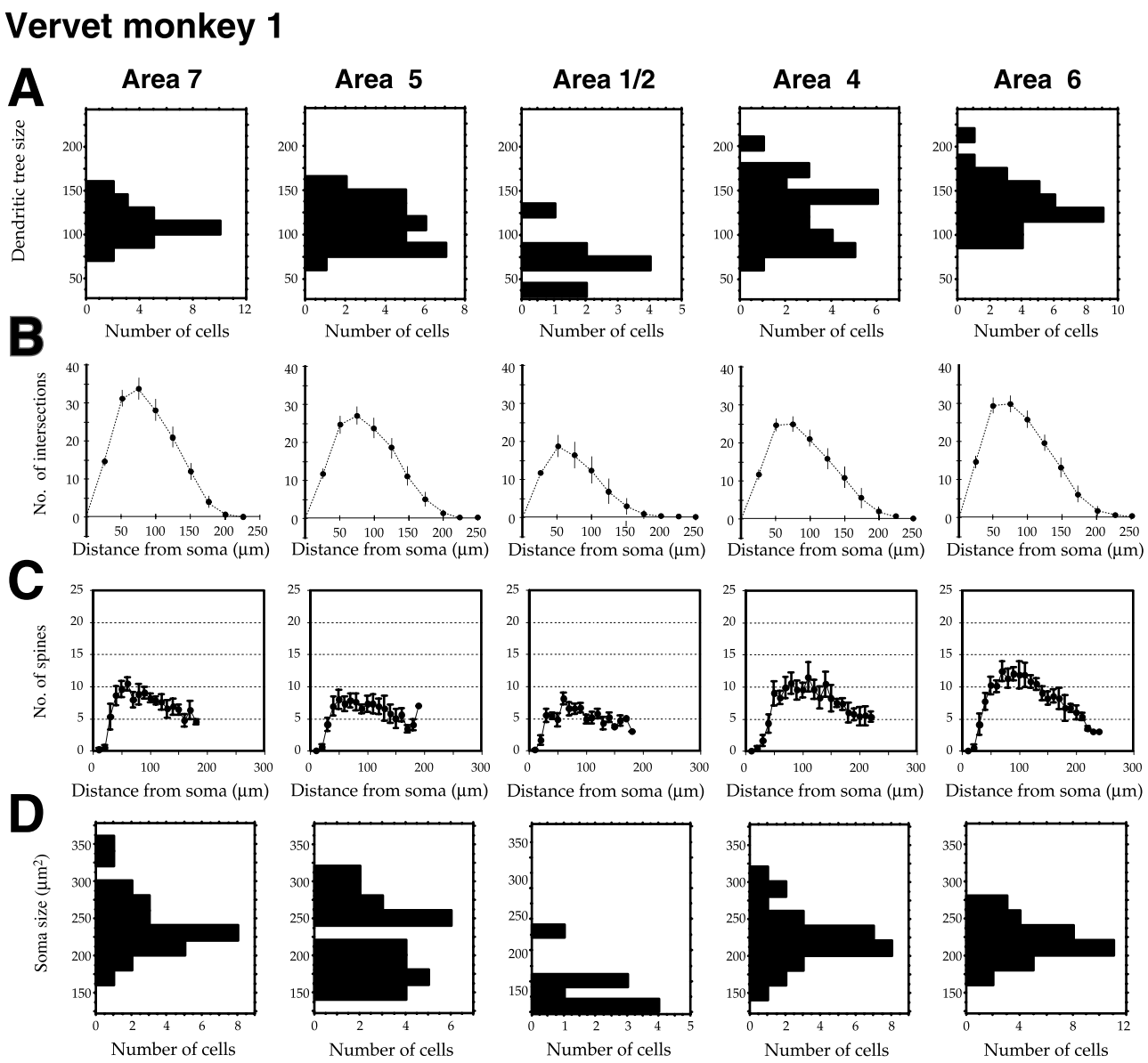



**Fig. 4.** Graphs illustrating (A) Frequency histograms of the size of, (B) profiles of the branching patterns of, and (C) plots of the number of dendritic spines along individual dendrites in the basal trees of layer III pyramidal neurones in areas 2, 1/2, 4, 6, and 7b of the vervet monkey 1. (D) Frequency histograms of the size of the cell bodies of the same pyramidal neurones in areas 2, 1/2, 4, 6, and 7 of vervet monkey 1. Error bars=standard errors.

body to the distal tips of the dendrites and then tapered with distal progression (Figs. 4C and 5C, Table 3). Several things become apparent from a quick comparison of the spine density data. First, the spine profiles differ between cortical areas; second, there is a trend for increasing spine density along the dendrites of cells in area 4 to area 6, and through somatosensory areas 1/2, 5 and 7 in both VM1 and VM2; third, spine density on the basal dendrites of pyramidal cells in area 4 was higher than that for cells in area 1/2. Repeated measures ANOVAs (cortical area×distance from soma×spine density), over the entire dendritic length, revealed a significant difference ($P<0.001$) in the distribution of spines between cells in sensory-motor cortex in both VM1 ($F_{(1,\ 4)}=18.3$) and VM2 ($F_{(1,\ 4)}=50.6$). Post hoc ANOVAs revealed mixed results. Spine density was significantly higher ($P<0.05$) in area 6 than area 4 in VM2 ($F_{(1,\ 1)}=91.58$), there were significant differences between somatosensory areas (VM1: 1/2 vs. 5, $F_{(1,\ 1)}=7.42$; VM2: 5 vs. 7, $F_{(1,\ 1)}=9.70$) and in VM1 there was a significant difference in spine density between cells in areas 1/2 and 4 ($F_{(1,\ 1)}=22.76$). However, there was no significant difference in spine density in the basal dendritic trees of cells between areas 6 and 4 in VM1 ($F_{(1,\ 1)}=1.32$, $P=0.27$), nor between areas 1/2 and 5 in VM2 ($F_{(1,\ 1)}=2.41$, $P=0.14$), nor areas 5 and 7 in VM1 ($F_{(1,\ 1)}=1.56$, $P=0.23$), nor areas 1/2 and 4 in VM2 ($F_{(1,\ 1)}=0.77$, $P=0.39$).

The total number of dendritic spines on the basal dendritic tree of the “average” pyramidal neurone in each cortical area was calculated by combining data from the Sholl analyses with that of spine densities (see Elston,

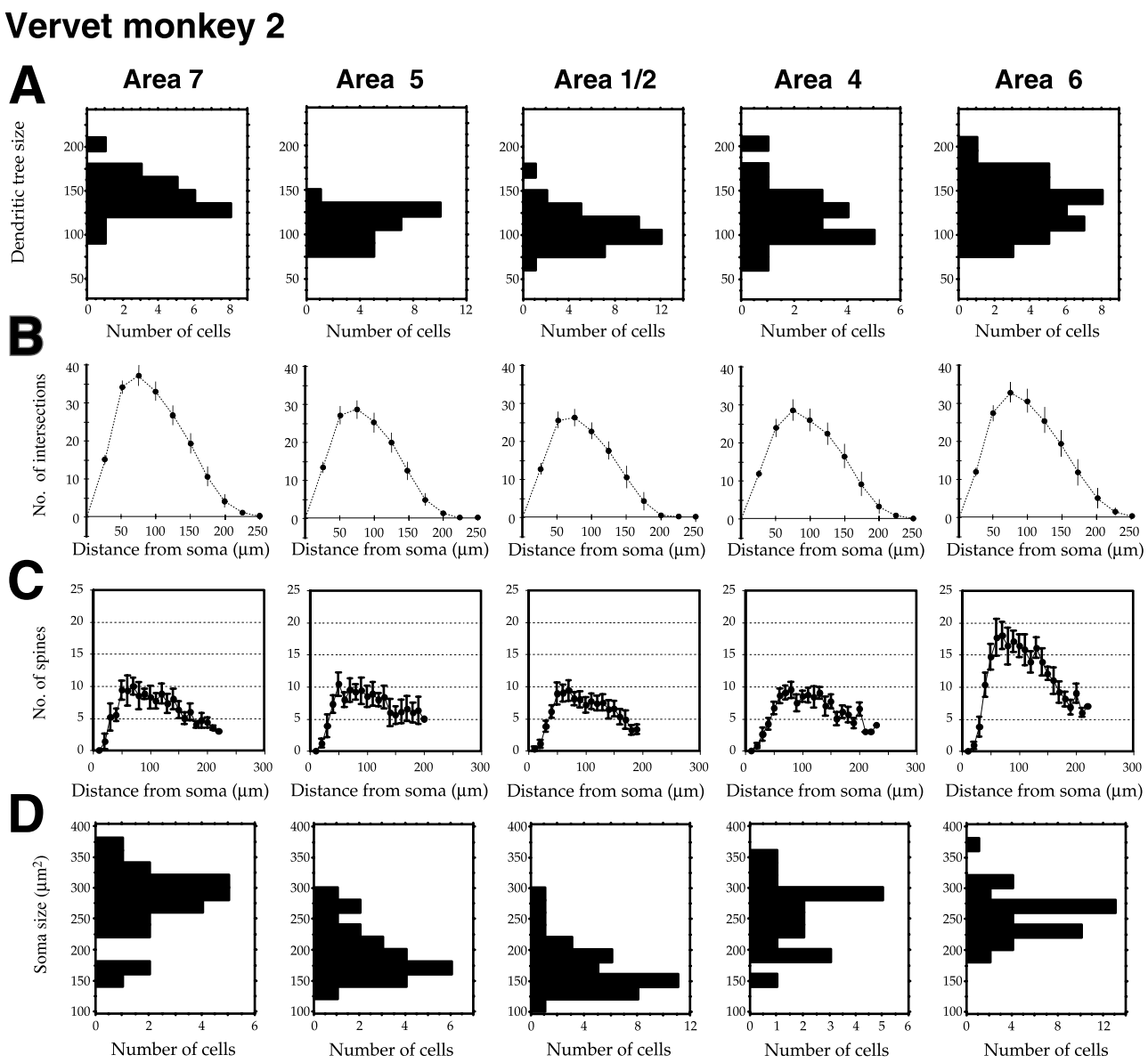



**Fig. 5.** Graphs illustrating (A) Frequency histograms of the size of, (B) profiles of the branching patterns of, and (C) plots of the number of dendritic spines along individual dendrites in the basal trees of layer III pyramidal neurones in areas 2, 1/2, 4, 6, and 7b of the vervet monkey 2. (D) Frequency histograms of the size of the cell bodies of the same pyramidal neurones in areas 2, 1/2, 4, 6, and 7 of vervet monkey 2. Error bars=standard errors.

2001). Based on our estimates, we found considerable differences in the number of spines in the basal dendritic tree of the "average" layer III pyramidal neurone in the different cortical areas (Table 4). Moreover, the differences were systematic between the different cortical areas: in both VM1 and VM2 cells in area 6 were more spinous than those in area 4, cells in area 7 were more spinous than those in area 5, cells in area 5 were more spinous than those in area 1/2 and cells in area 4 were more spinous than those in area 1/2.

### Somal areas

Cell bodies were drawn in the plane tangential to the cortical layers, their size calculated and frequency distributions plotted in Figs. 4D and 5D (see also Table 5). As can be seen from the figures, there was a degree of variation in the size of cell bodies between cortical areas in both VM1 and VM2. ANOVAs revealed these differences to be statistically significant ($P<0.001$) in both VM1 ($F_{(4)}=10.01$) and VM2 ($F_{(4)}=33.21$). Post hoc one way ANOVAs revealed no significant difference in the size of the somata between cells in areas 4 and 6 (VM1: $F_{(1)}=0.43$, $P=0.51$; VM2: $F_{(1)}=0.06$, $P=0.81$). In both cases there was a significant difference ($P<0.05$) in size of the somata between areas 1/2 and 5 (VM1: $F_{(1)}=17.09$; VM2: $F_{(1)}=4.54$). In VM2 there was a significant difference ($P<0.05$) in size of the cell bodies of neurons between areas 5 and 7 ($F_{(1)}=36.43$), but not in VM1 ($F_{(1)}=1.29$). In

**Table 2.** Results of Sholl analyses of the basal dendritic arbors of layer III pyramidal cells (mean±SEM) in sensorimotor areas 7, 5, 1/2, 4 and 6 in vervet monkey 1 (VM1) and vervet monkey 2 (VM2)

| VM1 | 25 μm | 50 μm | 75 μm | 100 μm | 125 μm | 150 μm | 175 μm | 200 μm | 225 μm | 250 μm | 275 μm | 300 μm |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 6 | 14.55±0.54 | 29.33±0.73 | **29.76**±0.75 | 25.70±0.79 | 19.51±0.73 | 13.03±0.88 | 5.94±0.74 | 1.61±0.45 | 0.42±0.22 | 0.91±0.09 | | |
| 4 | 11.54±0.51 | 24.68±0.66 | **24.93**±0.78 | 21.11±0.82 | 15.75±1.02 | 10.75±1.02 | 5.36±1.01 | 1.82±0.46 | 0.50±0.18 | 0.07±0.05 | 0.04±0.04 | |
| 1/2 | 11.56±0.34 | **18.67**±1.83 | 16.33±2.33 | 12.33±2.34 | 6.78±2.21 | 2.89±1.44 | 0.78±0.66 | 0.22±0.22 | 0.11±0.11 | | | |
| 5 | 11.65±0.43 | 24.61±0.83 | **26.97**±0.84 | 23.64±0.97 | 18.42±0.96 | 11.10±0.90 | 5.00±0.67 | 1.23±0.34 | 0.26±0.14 | 0.06±0.06 | | |
| 7 | 14.58±0.41 | 31.19±0.82 | **33.69**±1.16 | 27.92±1.16 | 21.12±1.09 | 12.04±0.90 | 4.00±0.58 | 0.69±0.20 | 0.04±0.04 | | | |
| VM2 | 25 μm | 50 μm | 75 μm | 100 μm | 125 μm | 150 μm | 175 μm | 200 μm | 225 μm | 250 μm | 275 μm | 300 μm |
| 6 | 11.85±0.31 | 27.32±0.64 | **32.83**±0.79 | 30.51±0.97 | 25.41±1.04 | 19.31±1.10 | 11.83±1.08 | 4.88±0.78 | 1.24±0.30 | 0.12±0.08 | 0.02±0.02 | |
| 4 | 11.71±0.43 | 23.90±0.99 | **28.67**±1.17 | 26.10±1.26 | 22.38±1.23 | 16.48±1.41 | 9.09±1.36 | 3.19±0.83 | 0.76±0.36 | 0.19±0.09 | 0.05±0.05 | |
| 1/2 | 12.66±0.43 | 25.58±0.71 | **26.26**±0.68 | 22.68±0.72 | 17.58±0.76 | 10.47±0.91 | 4.13±0.74 | 0.45±0.18 | | | | |
| 5 | 13.32±0.54 | 27.11±0.91 | **28.64**±0.92 | 25.18±0.99 | 19.82±0.97 | 12.43±0.88 | 4.75±0.63 | 1.14±0.27 | 0.11±0.06 | | | |
| 7 | 14.84±0.45 | 34.00±0.76 | **37.12**±1.08 | 33.00±0.99 | 26.56±1.03 | 19.12±1.05 | 10.40±1.00 | 3.88±0.67 | 0.80±0.34 | 0.16±0.09 | | |

both VM1 and VM2 there was a significant difference in the size of the cell bodies of pyramidal cells between areas 1/2 and 4 ($F_{(1)}$=35.14 and $F_{(1)}$=46.97, respectively).

## DISCUSSION

This study forms part of an ongoing investigation into specializations in pyramidal cell structure in the cerebral cortex of primates. Here we focused on sensory-motor cortex in the vervet monkey (savannah guenon). The vervet monkey was chosen for study for several reasons; (1) according to Brodmann (1913), despite having a smaller total cortical surface area, and unlike the trend reported in visual cortex, motor cortex is larger in guenons than in the macaque monkey, (2) motor cortex occupies a disproportionately larger portion of the frontal lobe in the vervet monkey compared with the macaque monkey and (3) maps often reproduced from Brodmann's works that are labeled as the macaque monkey are reportedly those of the guenon (Fig. 6, Table 6). The study cell structure in sensory-motor cortex of closely related monkeys in which these cortical areas have undergone different degrees of cortical expansion may reveal new insights into the evolution of specialized pyramidal cell phenotypes in the primate cerebral cortex.

### Specialization in the pyramidal cell phenotype in primate sensory-motor cortex

As in our previous study in the macaque monkey (Elston and Rockland, 2002), we found differences in pyramidal cell structure between various cortical areas that contain neurons involved somatosensory and motor function. In addition, the differences in pyramidal cell structure were systematic in the vervet monkey, and followed the same basic trends as reported in the macaque monkey. In particular, the basal dendritic trees of layer III pyramidal cells were progressively larger, more branched and more spinous with progression through somatosensory areas 1/2, 5 and 7, as well as from the primary motor area (Brodmann's area 4) to premotor cortex (Brodmann's area 6). Interestingly, however, the extent of specialization in pyramidal cell structure in sensory-motor cortex varied between species (Fig. 7). In particular, the extent to which the size of the basal dendritic trees of pyramidal cells increased with progression from area 1/2 to area 5 and area 7 was less in the vervet monkey than in the macaque monkey. The increase in branching complexity and spine density in the basal trees with progression through these same areas was also less notable in the vervet monkey than in the macaque monkey.

It may be argued that these differences in the extent of specialization in pyramidal cell structure in sensory-motor cortex of the vervet monkey and macaque monkey can be attributed to methodological error. For example, as dendritic atrophy and spine loss occurs during aging (Scheibel et al., 1975, 1976; Huttenlocher, 1979; Nakamura et al., 1985; Jacobs and Scheibel, 1993; Jacobs et al., 1993, 1997; Anderson and Rutledge, 1996; Huttenlocher and Dabholkar, 1997), differences in cell morphology reported

**Table 3.** Average spine density per 10 μm along the basal dendrites of layer III pyramidal cells as a function of distance from the cell bodies to the distal dendritic tips (mean±SEM, maximum in bold)

| VM1 | 0–10 μm | 11–20 μm | 21–30 μm | 31–40 μm | 41–50 μm | 51–60 μm | 61–70 μm | 71–80 μm | 81–90 μm | 91–100 μm |
|---|---|---|---|---|---|---|---|---|---|---|
| 7 | 0.20±0.13 | 0.50±0.22 | 5.30±1.24 | 8.60±0.96 | 9.60±0.78 | **10.4**±0.64 | 7.90±0.78 | 8.80±1.01 | 9.00±0.56 | 8.20±0.44 |
| 5 | 0 | 0.60±0.27 | 4.10±0.64 | 6.90±0.95 | **8.00**±0.97 | 7.30±0.72 | 7.80±0.71 | 7.60±0.82 | 6.50±0.40 | 7.30±0.82 |
| 1/2 | 0.10±0.10 | 1.60±0.48 | 5.50±0.65 | 5.40±0.34 | 4.80±0.66 | **8.10**±0.57 | 6.50±0.65 | 6.50±0.50 | 6.70±0.50 | 4.90±0.43 |
| 4 | 0 | 0.40±0.22 | 1.60±0.56 | 4.30±0.90 | 9.00±1.20 | 8.30±0.67 | 9.90±1.08 | 10.6±1.07 | 9.50±0.96 | 9.50±0.72 |
| 6 | 0 | 0.50±0.31 | 4.10±1.05 | 7.70±0.82 | 10.2±0.85 | 10.1±0.59 | **12.4**±0.95 | 11.3±0.94 | 12.0±0.63 | 11.8±1.35 |

| VM1 | 101–110 μm | 111–120 μm | 121–130 μm | 131–140 μm | 141–150 μm | 151–160 μm | 161–170 μm | 171–180 μm | 181–190 μm | 191–200 μm |
|---|---|---|---|---|---|---|---|---|---|---|
| 7 | 7.80±0.39 | 7.60±0.76 | 6.50±0.87 | 5.50±1.15 | 4.50±1.07 | 1.90±0.85 | 1.90±1.06 | 0.90±0.60 | 0 | 0 |
| 5 | 7.30±0.91 | 6.90±0.74 | 6.60±1.24 | 5.20±1.04 | 4.50±1.04 | 3.90±1.00 | 1.70±0.62 | 1.20±0.66 | 0.70±0.70 | 0 |
| 1/2 | 5.10±0.57 | 4.60±0.86 | 3.40±0.79 | 3.10±0.91 | 1.50±0.62 | 1.40±0.75 | 0.50±0.50 | 0.30±0.30 | 0 | 0 |
| 4 | **11.5**±1.45 | 9.60±0.97 | 8.30±1.10 | 10.4±1.23 | 7.40±1.47 | 6.70±0.83 | 6.60±0.92 | 4.80±1.18 | 4.00±0.99 | 3.80±1.11 |
| 6 | 11.8±1.20 | 10.8±0.61 | 10.4±0.58 | 9.00±0.70 | 8.20±0.65 | 7.70±1.07 | 6.50±1.43 | 4.00±1.45 | 4.00±1.13 | 2.40±1.06 |

| VM1 | 201–210 μm | 211–220 μm | 221–230 μm | 231–240 μm | 241–250 μm | 251–260 μm | 261–270 μm | 271–280 μm | 281–290 μm | 291–300 μm |
|---|---|---|---|---|---|---|---|---|---|---|
| 7 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 5 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 1/2 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 4 | 2.20±1.05 | 1.60±0.85 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 6 | 1.60±0.83 | 0.70±0.47 | 0.30±0.30 | 0.30±0.30 | 0 | 0 | 0 | 0 | 0 | 0 |

| VM2 | 0–10 μm | 11–20 μm | 21–30 μm | 31–40 μm | 41–50 μm | 51–60 μm | 61–70 μm | 71–80 μm | 81–90 μm | 91–100 μm |
|---|---|---|---|---|---|---|---|---|---|---|
| 7 | 0 | 1.40±0.75 | 5.20±1.36 | 5.50±0.62 | 9.40±0.96 | 9.30±1.48 | **10.00**±0.82 | 8.50±1.31 | 8.80±0.53 | 8.30±1.14 |
| 5 | 0 | 1.20±0.44 | 3.90±1.11 | 7.30±0.87 | **10.40**±1.14 | 8.00±0.75 | 9.50±1.18 | 9.20±0.74 | 9.40±1.30 | 8.50±1.05 |
| 1/2 | 0.30±0.21 | 1.10±0.41 | 3.80±0.51 | 6.10±0.62 | 8.90±1.10 | 9.00±0.83 | **9.40**±1.00 | 8.10±0.66 | 8.00±0.82 | 7.10±0.77 |
| 4 | 0 | 0.80±0.25 | 2.60±0.67 | 4.20±0.59 | 6.70±0.62 | 8.60±0.73 | 9.10±0.67 | **9.50**±0.78 | 7.50±0.82 | 8.50±0.56 |
| 6 | 0 | 0.90±0.38 | 3.80±0.99 | 10.30±1.20 | 14.70±1.30 | 17.70±1.80 | **18.00**±1.36 | 16.40±1.81 | 17.10±1.03 | 16.40±1.14 |

| VM2 | 101–110 μm | 111–120 μm | 121–130 μm | 131–140 μm | 141–150 μm | 151–160 μm | 161–170 μm | 171–180 μm | 181–190 μm | 191–200 μm |
|---|---|---|---|---|---|---|---|---|---|---|
| 7 | 7.70±0.79 | 8.80±0.99 | 7.30±0.93 | 8.00±1.02 | 6.30±0.83 | 4.50±0.76 | 4.80±1.08 | 2.00±0.70 | 1.40±0.79 | 1.30±0.70 |
| 5 | 8.90±1.19 | 8.00±0.65 | 8.40±1.16 | 6.00±1.32 | 5.60±0.87 | 4.90±1.31 | 3.30±1.37 | 2.40±1.14 | 1.90±1.15 | 0.50±0.50 |
| 1/2 | 7.70±0.78 | 7.40±0.72 | 7.50±0.81 | 6.40±0.91 | 6.60±0.76 | 4.80±0.88 | 3.40±1.05 | 1.60±0.62 | 1.00±0.56 | 0 |
| 4 | 8.70±0.70 | 8.40±1.01 | 9.00±0.56 | 7.00±0.93 | 7.70±0.68 | 4.50±0.81 | 5.50±0.85 | 3.40±1.09 | 2.60±0.79 | 1.30±0.90 |
| 6 | 15.80±1.49 | 13.90±1.06 | 16.10±1.00 | 13.90±1.31 | 12.10±0.66 | 11.00±1.32 | 8.30±1.34 | 6.50±1.25 | 4.10±1.24 | 3.60±1.56 |

| VM2 | 201–210 μm | 211–220 μm | 221–230 μm | 231–240 μm | 241–250 μm | 251–260 μm | 261–270 μm | 271–280 μm | 281–290 μm | 291–300 μm |
|---|---|---|---|---|---|---|---|---|---|---|
| 7 | 0.70±0.47 | 0.30±0.30 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 5 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 1/2 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 4 | 0.30±0.30 | 0.30±0.30 | 0.40±0.40 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 6 | 1.20±0.81 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

here between the vervet monkey and macaque may reflect possible differences in the age of the animals selected for study. Alternatively, it could be argued that the results reflect interindividual variation. While we cannot discount these possibilities unequivocally, our results show that species differences reported here in sensory-motor cortex are not present in cingulate or visual cortex of the same cerebral hemispheres of the same animals (Elston et al., 1999b, 2005; Elston et al., in preparation). In addition, our unpublished studies or layer III pyramidal cells in sensory-

**Table 4.** Estimates of the total number of spines in the basal dendritic tree of the "average" layer III pyramidal cell in sensory-motor cortex of the vervet monkey

| Case | 7 | 5 | 1/2 | 4 | 6 |
|---|---|---|---|---|---|
| VM1 | 2715 | 1904 | 906 | 2242 | 3201 |
| VM2 | 3295 | 2477 | 2055 | 2500 | 5499 |

Note the variation reported in area 1/2 between the two cases. Data were sampled more anteriorly in VM1, probably corresponding to area 1, and more posteriorly in VM2, probably corresponding to area 2 (see Fig. 2).

**Table 5.** Size of the somata of layer III pyramidal cells (in the tangential plane) in cortical areas 7, 5, 1/2, 4 and 6 in vervet monkey 1 (VM1) and vervet monkey 2 (VM2)

| Visual area | *n* | Mean $\mu m^2$ | S.D. | S.E.M. | Minimum $\mu m^2$ | Maximum $\mu m^2$ |
|---|---|---|---|---|---|---|
| VM1 | | | | | | |
| 7 | 26 | 240.04 | 41.38 | 8.12 | 178.62 | 346.25 |
| 5 | 31 | 224.47 | 58.77 | 10.56 | 143.26 | 402.08 |
| 1/2 | 9 | 137.96 | 39.45 | 13.15 | 100.40 | 225.94 |
| 4 | 28 | 223.26 | 36.97 | 6.99 | 153.16 | 309.58 |
| 6 | 33 | 217.85 | 27.22 | 4.74 | 166.33 | 271.29 |
| VM2 | | | | | | |
| 7 | 25 | 277.92 | 53.55 | 10.71 | 149.82 | 374.22 |
| 5 | 24 | 193.79 | 43.23 | 8.82 | 127.69 | 286.97 |
| 1/2 | 38 | 170.71 | 40.46 | 6.56 | 102.60 | 282.66 |
| 4 | 19 | 257.36 | 53.12 | 12.19 | 143.55 | 343.82 |
| 6 | 40 | 254.55 | 36.64 | 5.79 | 191.43 | 370.63 |

motor, cingulated and visual cortex in the baboon confirm this trend (Elston et al., in preparation; see Elston, in press for a review). Thus, it is more likely that the relative differences in the extent of specialization in the pyramidal cell phenotype in sensory-motor areas of these primates reflect topographic variation within sensory-motor cortex rather than species specializations or sampling error.

### Functional implications of cell specialization

The data from the vervet monkey and macaque monkey reveal a consistent trend for a systematic increase in the number of spines in the dendritic trees of cells with progression through areas 1/2, 5 and 7, as well as from area 4 to area 6, is likely to be important for specialist functions performed by neurones in these cortical areas. We propose that these differences in neuron structure impart different computational abilities in their respective circuitry. Moreover, differences in pyramidal cell structure, and the circuits they comprise, in homologous cortical areas between species may subserve their different functional/behavioral capabilities. These possibilities have been discussed in length elsewhere (Elston, 2002, 2003a; Jacobs and Scheibel, 2002; Chklovskii et al., 2004) and are detailed briefly below.

As pyramidal cells comprise 70–90% of all neurones in the cerebral cortex, form extensive connective networks within cortical areas, make nearly all connections between cortical areas (both ipsi- and contralateral) and contain the excitatory transmitter glutamate, differences in their structure are likely to influence patterns of connectivity in the circuits they comprise both within and between cortical areas. These differences in neuronal structure and the patterns of connectivity are likely to influence the computational power at both the cellular and systems levels (see Jacobs and Scheibel, 2002; Elston, 2003a for reviews). At the cellular level, the size of the dendritic trees may influ-

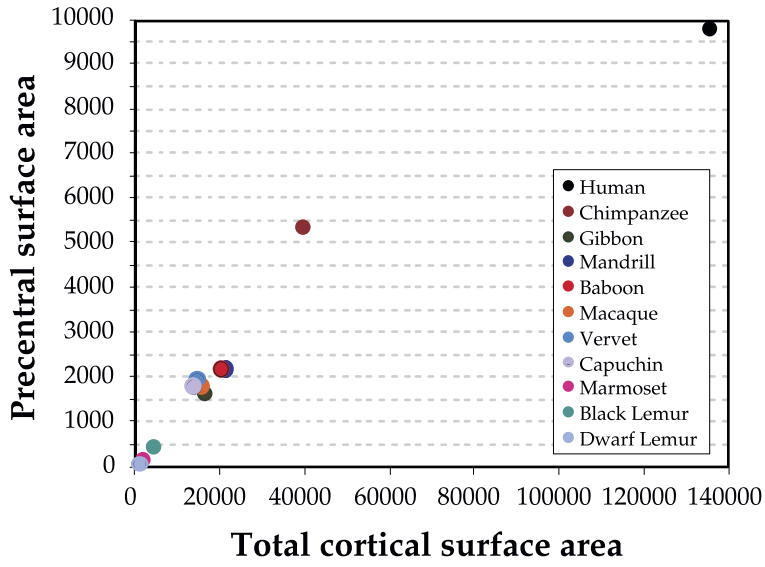



**Fig. 6.** Graph illustrating the relationship between the total cortical surface area vs. the cortical surface area of precentral cortex (defined as areas 4 and 6 by Brodmann). Linear regression analysis revealed a close correlation between the two variables ($y=0.0717x+725.47$, $r^2=0.93$). Data on both axes are $mm^2$. Data taken from Brodmann (1909, 1913), see Garey (1994) and Elston and Garey (2004) for translations of original works.

**Table 6.** Data on the total surface area of the cerebral cortex, motor cortex and visual cortex originally reported by Brodmann (1913)

| | Total | Motor | Visual |
|---|---|---|---|
| Human | 135,470 | 9833 | 3495 |
| Chimpanzee | 39,572 | 5389 | 3213 |
| Gibbon | 16,301 | 1651 | 1899 |
| Mandrill | 21,321 | 2194 | 3537 |
| Baboon (*P. hamadryas*) | 20,594 | 2898 | 2130 |
| Baboon (*P. cynocephalus*) | 20,376 | 2200 | 2559 |
| Macaque | 15,308 | 1817 | 1866 |
| Guenon | 14,641 | 1976 | 2210 |
| Capuchin monkey | 13,682 | 1822 | 1786 |
| Marmoset | 1649 | 167 | 341 |
| Black lemur | 4054 | 453 | 440 |
| Dwarf lemur | 921 | 94 | 132 |

ence their sampling geometry, larger arbors being able to sample from a larger cross-sections of cortex than smaller arbors. In sensory-motor cortex this translates into a larger portion of the body. Differences in the number of dendritic branches may influence the propagation of potentials throughout the trees (see Stuart et al., 1997; Koch, 1999; Mel, 1999; Spruston et al., 1999; Häusser et al., 2000 for reviews) and the representational power of the cells (Poirazi and Mel, 2001). Differences in the number of spines (putative excitatory inputs) may reflect difference in the number of neurones from which they receive inputs. At the systems level, circuits composed of neurones that are able to receive large numbers of inputs may integrate a larger number of more diverse synapses that those in circuits composed of neurones that receive fewer inputs. Thus, cells in areas 7, for example, may be characterized by a higher degree of connectivity with other cortical areas than those in area 1/2. This ability to integrate large numbers of inputs from diverse cortical areas may be important for binding information across functionally related ensembles of neurones in hierarchical or distributed systems (see Felleman and Van Essen, 1991; Mountcastle, 1995; Singer, 1998; Friston, 2001 for reviews). In addition, circuits characterized by complex patterns of connectivity may have greater potential for plastic change than those composed of less complex patterns of connectivity (Stepanyants et al., 2002).

### Pyramidal cell specialization in evolution

Studies in the primate cerebral cortex reveal two different trends in the evolution of the pyramidal cell phenotype during cortical expansion: (1) expansion whereby mature cortex contains pyramidal neurones of similar structure; and (2) expansion whereby mature cortex contains pyramidal neurones of more complex structure. For example, despite a five-fold difference the total cortical surface area (Brodmann, 1913 see Elston and Garey, 2004 for a translation), layer III pyramidal cells in V1 of macaques and marmosets have remarkably similar number of spines in their basal dendritic trees (Elston and Rosa, 1997, 1998; Elston et al., 1999a). On the contrary, pyramidal cells in inferotemporal cortex and granular prefrontal cortex of primates are larger, more branched and more spinous in the brains of species in which these regions are volumetrically larger (Elston et al., 1999b, 2001; Elston et al., in preparation; Elston, 2003b; see Elston, 2003c for a review). Studies of pyramidal cells in the cerebral cortex of non-primates (Elston et al., 2004) reveal at least three additional possibilities for trends in pyramidal cell specialization in evolution: an increase in complexity in the structure of pyramidal cells with little or no cortical expansion; an increase in complexity in the structure of pyramidal cells during reduction in cortical size; and a decrease in complexity in the structure of pyramidal cells during cortical expansion (see Elston, 2005). Based on presently available data it remains unclear which of these scenarios may have occurred in primate sensory-motor cortex. However, based on Brodmann's data (Brodmann, 1913) that show considerable expansion of motor cortex in primates (Fig. 6), the first two of these scenarios are unlikely to apply. Thus we are left with three possibilities related to the nature of pyramidal cell specialization during the evolution of sensory-motor cortex in primates. Clearly, further comparative studies are required to extend the basis for comparison.

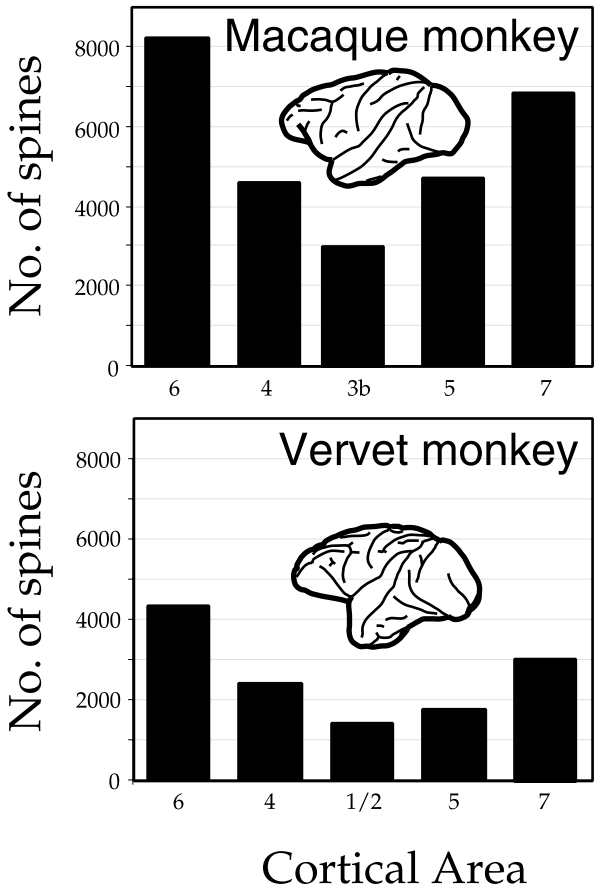



**Fig. 7.** Bar graphs illustrating our estimates of the total number of spines in the basal dendritic tree of the "average" layer III pyramidal cell in motor (4 and 6) and somatosensory (1/2, 5 and 7) cortex in the (A) vervet monkey and (B) macaque monkey.

## CONCLUSIONS

Here we have demonstrated variation in the pyramidal cell structure in sensory-motor cortex of the vervet monkey. The trends in pyramidal cell specialization observed here are similar to, but less in extent, than those reported in the macaque monkey. Selective pressures driving these specializations in somatosensory and motor cortex remain unclear, but it appears as though, as in visual cortex, there are two trends for specialization: one for constancy in neuron structure and the other for increasing complexity. In future studies it will be worthwhile to study pyramidal cell structure in sensory-motor cortex of other primate species. In addition, it will be worthwhile to study pyramidal cells in other cortical layers, as well as the apical dendrites, in a variety of species. Only then will features that shape the mature pyramidal cell phenotype in somatosensory/motor cortex become more apparent.

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